Physics 4220H: Assignment III

Jeremy Favro (0805980)
Trent University, Peterborough, ON, Canada

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Problem 1. Starting from the following expression for (the square of) the complex propagation constant in a conductor,

$$\widetilde{k}^2 = \mu\epsilon\omega^2 + i\mu\sigma\omega = \mu\omega\left[\epsilon\omega + i\sigma\right]$$

Go through the algebra to find expressions for the real ($\|$) and imaginary (κ) parts of the propagation constant. i.e. find $\|$ and κ, where $\widetilde{k} = \| + \kappa$.

Solution 1. To do this I will write $\widetilde{k}$ in polar form, apply the square root, then use the half-angle formulae

$$\cos(x/2) = \sqrt{\frac{1+\cos(x)}{2}} \quad \sin(x/2) = \sqrt{\frac{1-\cos(x)}{2}}$$

and the formulae (derivable by considering a triangle with sides 1, x and $\sqrt{1+x^2}$)

$$\cos(\arctan(x)) = \frac{1}{\sqrt{1+x^2}} \quad \sin(\arctan(x)) = \frac{x}{\sqrt{1+x^2}}.$$

To start, we can write

$$\widetilde{k}^2 = Ke^{i\phi}$$

with

$$K = \left|\widetilde{k}\right| = \mu\omega\sqrt{(\epsilon\omega)^2+\sigma^2} \quad \phi = \arctan\Big(\frac{\sigma}{\epsilon\omega}\Big).$$

We then write

$$\widetilde{k} = \sqrt{Ke^{i\phi}} = \sqrt{K}e^{i\phi/2} = \sqrt{\mu\omega\sqrt{(\epsilon\omega)^2+\sigma^2}}\left[\cos\left(\frac{1}{2}\arctan\Big(\frac{\sigma}{\epsilon\omega}\Big)\right) + i\sin\left(\frac{1}{2}\arctan\Big(\frac{\sigma}{\epsilon\omega}\Big)\right)\right]$$

$$= \omega\sqrt{\mu\epsilon}\sqrt{\sqrt{1+\Big(\frac{\sigma}{\epsilon\omega}\Big)^2}}\left[\cos\left(\frac{1}{2}\arctan\Big(\frac{\sigma}{\epsilon\omega}\Big)\right) + i\sin\left(\frac{1}{2}\arctan\Big(\frac{\sigma}{\epsilon\omega}\Big)\right)\right]$$

and then use the half-angle identities to write this as

$$\widetilde{k} = \omega\sqrt{\frac{\mu\epsilon}{2}}\sqrt{\sqrt{1+\Big(\frac{\sigma}{\epsilon\omega}\Big)^2}}\left[\sqrt{1+\cos\Big(\arctan\Big(\frac{\sigma}{\epsilon\omega}\Big)\Big)} + i\sqrt{1-\cos\Big(\arctan\Big(\frac{\sigma}{\epsilon\omega}\Big)\Big)}\right]$$

and then apply the $\cos(\arctan(x))$ identity to rewrite that as

$$\widetilde{k} = \omega\sqrt{\frac{\mu\epsilon}{2}}\sqrt{\sqrt{1+\Big(\frac{\sigma}{\epsilon\omega}\Big)^2}}\left[\sqrt{1+\frac{1}{\sqrt{1+\left(\frac{\sigma}{\epsilon\omega}\right)^2}}} + i\sqrt{1-\frac{1}{\sqrt{1+\left(\frac{\sigma}{\epsilon\omega}\right)^2}}}\right].$$

Wow! Now multiply in the square root out front,

$$\widetilde{k} = \omega\sqrt{\frac{\mu\epsilon}{2}}\left[\sqrt{\sqrt{1+\Big(\frac{\sigma}{\epsilon\omega}\Big)^2}+1} + i\sqrt{\sqrt{1+\Big(\frac{\sigma}{\epsilon\omega}\Big)^2}-1}\right].$$

so

$$\| = \omega\sqrt{\frac{\mu\epsilon}{2}}\sqrt{\sqrt{1+\left(\frac{\sigma}{\epsilon\omega}\right)^2}+1}$$

and

$$\kappa = \omega\sqrt{\frac{\mu\epsilon}{2}}\sqrt{\sqrt{1+\left(\frac{\sigma}{\epsilon\omega}\right)^2}-1}.$$

Problem 2. Household microwave ovens operate at a frequency of $\nu = 2.54\,\text{GHz}$. Imagine that you leave a $20\,^\circ\text{C}$ stainless-steel spoon ($s = 1.5\times10^6\,\text{S}\,\text{m}^{-1}$, $\mu \approx \mu_0$, $\epsilon \approx \epsilon_0$) next to a bowl of soup in the microwave.

(a) Estimate the proportion of the incident field that is reflected and transmitted from/into the spoon.

(b) Estimate the skin-depth over which the microwaves that enter the handle are absorbed

(c) Imagine that the full power of a 1 kW microwave is incident onto the spoon's handle, after which any reflected power gets absorbed by the soup. If the spoon weighs 30 g and the steel has a heat capacity of $450\,\text{J}\,\text{kg}^{-1}\,\text{K}^{-1}$, what temperature do you expect the spoon to be at the end of a 2-minute full-power heating cycle?

Solution 2.

(a) I have no idea how egregious I can be with my approximations, so I'm going to lean pretty heavy towards spherical cow. Assuming the spoon is a flat surface (i.e. a rectangular block of steel) and the microwaves are normally incident on any given side of the spoon, the reflection and transmission coefficients are

$$r = \frac{n_{air}-n_{steel}}{n_{air}+n_{steel}} \quad t = \frac{2n_{air}}{n_{air}+n_{steel}}.$$

We can ballpark $n_{air} \approx 1$ (I found it to be around 0.995) and obtain n_{steel} using our expression for $\widetilde{k}$. The expressions for $\|$ and κ are very similar since $\sigma \gg 1$. Approximating the two as equal and plugging in numbers, noting that $\omega = 2\pi\nu$ we obtain $\widetilde{k} \approx 1.2\times10^5\,\text{m}^{-1}(1+i)$ and hence $\widetilde{n}_{steel} = c\widetilde{k}/\omega \approx 2.25\times10^3(1+i)$. Plugging these into the field coefficients we get that $r \approx -0.999$ and $t \approx 8.8\times10^{-4}$. This lines up at least a bit with what I would expect a metal to do with microwaves, mostly reflection.

(b) Using the same numbers from above, $d = 1/\kappa \approx 8.3\,\mu\text{m}$.

(c) Neglecting any of the fancy stuff that microwaves do to heat food evenly, and assuming all of the transmitted power is absorbed we apply $T = 1-R = 1-r^2 = 1.77\times10^{-3}$. We assume that the spoon sees all of the 1 kW and so absorbs $1.77\times10^{-3}\,\text{kW} = 1.77\,\text{W}$. Since the spoon weighs 30 g it takes 13.5 J to increase the temperature by one Kelvin. Then since then spoon is absorbing $1.77\,\text{J}\,\text{s}^{-1}$ for 120 s, the total absorbed energy is 212.4 J and so the increase in temperature is 15.73 K, which amounts to an end temperature of $35.73\,^\circ\text{C}$, which seems really low?

Does this work? Intensity is W/m^2

Problem 3. Consider an isolated resonance in a dielectric material for which the Lorentz Oscillator Model holds well. Using the dispersion expression for the complex index of refraction show that the anomalous dispersion regime (in n) is concurrent with the span of the full-width at half-maximum (FWHM) of the absorption profile (α).

Solution 3. The complex refractive index is

$$\widetilde{n} = \left[1+\frac{Nq^2}{2m\epsilon_0}\sum_j \frac{f_j(\omega_j^2-\omega^2)}{(\omega_j^2-\omega^2)^2+\gamma_j^2\omega^2}\right] + i\frac{Nq^2}{m\epsilon_0 c}\sum_j \frac{f_j\gamma_j\omega^2}{(\omega_j^2-\omega^2)^2+\gamma_j^2\omega^2}$$

where we call the real part n and the imaginary part α. The anomalous dispersion regime is the region of $n(\omega)$ for which $dn/d\omega < 0$. So, to find the "borders" of this region, we consider $dn/d\omega = 0$. To simplify things, since we are working with an ω_j very far from other ω_js, we consider only the jth term of the sum, as the other terms are very small. Then,

$$\begin{aligned}\frac{dn}{d\omega} &= \frac{Nq^2f_j}{2m\epsilon_0}\frac{d}{d\omega}\left[\frac{(\omega_j^2-\omega^2)}{(\omega_j^2-\omega^2)^2+\gamma_j^2\omega^2}\right] \\ &= \frac{Nq^2f_j}{2m\epsilon_0}\frac{(-2\omega)((\omega_j^2-\omega^2)^2+\gamma_j^2\omega^2)-(2(-2\omega)(\omega_j^2-\omega^2)+2\gamma_j^2\omega)(\omega_j^2-\omega^2)}{\left[(\omega_j^2-\omega^2)^2+\gamma_j^2\omega^2\right]^2}\end{aligned}$$

we then set this to zero to find where the derivative changes sign

$$0 = \frac{\cancel{Nq^2 f_j}}{\cancel{2}m\epsilon_0}\frac{(-\cancel{2}\omega)((\omega_j^2-\omega^2)^2+\gamma_j^2\omega^2)-(2(-\cancel{2}\omega)(\omega_j^2-\omega^2)+\cancel{2}\omega\gamma_j^2)(\omega_j^2-\omega^2)}{\left[(\omega_j^2-\omega^2)^2+\gamma_j^2\omega^2\right]^2} \qquad \text{Disregarding the } \omega=0 \text{ solution}$$
$$= \frac{((\omega_j^2-\omega^2)^2+\gamma_j^2\omega^2)+(\gamma_j^2-2(\omega_j^2-\omega^2))(\omega_j^2-\omega^2)}{\left[(\omega_j^2-\omega^2)^2+\gamma_j^2\omega^2\right]^2}$$

and we consider only the numerator since the denominator will never blow up

$$\begin{aligned}
0 &= ((\omega_j^2-\omega^2)^2+\gamma_j^2\omega^2)+(\gamma_j^2-2(\omega_j^2-\omega^2))(\omega_j^2-\omega^2)\\
&= (\omega_j^2-\omega^2)^2-2(\omega_j^2-\omega^2)^2+\gamma_j^2\omega^2+\gamma_j^2(\omega_j^2-\omega^2)\\
&= -(\omega_j^2-\omega^2)^2+\cancel{\gamma_j^2\omega^2}+\gamma_j^2(\omega_j^2-\cancel{\omega^2})\\
\implies 0 &= (\omega_j^2-\omega^2)^2-\gamma_j^2\omega_j^2\\
\implies \omega &= \sqrt{\omega_j^2\pm\omega_j\gamma_j}
\end{aligned}$$

where I've introduced a $\pm$ inside the square root to obtain two solutions (looking at a graph to justify it) but I've chosen the principle square root outside. Why? I don't really know? If I had taken $\pm$ outside as well, the negative frequency components are symmetric with the positive ones anyway, so maybe that's it. Things work out anyway.

Now, far from other resonances, the maximum value of α occurs at the resonant frequency, $\omega=\omega_j$. This maximum value is then

$$\alpha_{max} = \frac{Nq^2 f_j}{m\epsilon_0 c}\frac{1}{\gamma_j}.$$

The half-max value then is $\alpha_{max}/2$. We then solve for ω when $\alpha=\alpha_{max}/2$,

$$\begin{aligned}
\frac{\cancel{Nq^2 f_j}}{\cancel{m\epsilon_0 c}}\frac{1}{2\gamma_j} &= \frac{\cancel{Nq^2 f_j}}{\cancel{m\epsilon_0 c}}\frac{\gamma_j\omega^2}{(\omega_j^2-\omega^2)^2+\gamma_j^2\omega^2}\\
\implies 2\gamma_j^2\omega^2 &= (\omega_j^2-\omega^2)^2+\gamma_j^2\omega^2\\
\implies 0 &= \omega^4-(2\omega_j^2+\gamma_j^2)\omega^2+\omega_j^4
\end{aligned}$$

which is quadratic in ω^2 so

$$\omega^2 = \frac{(2\omega_j^2+\gamma_j^2)\pm\sqrt{(2\omega_j^2+\gamma_j^2)^2-4\omega_j^4}}{2}$$

and hence

$$\omega = \sqrt{\frac{(2\omega_j^2+\gamma_j^2)\pm\gamma_j\sqrt{4\omega_j^2+\gamma_j^2}}{2}}.$$

In the limit that $\omega_j \gg \gamma_j$, $2\omega_j^2+\gamma_j^2 \approx 2\omega_j^2$ and we recover

$$\omega = \sqrt{\omega_j^2\pm\omega_j\gamma_j},$$

so the half-height points of the absorption peak indeed lie at the bounds of the anomalous dispersion region for resonances far from other resonances when $\omega_j \gg \gamma_j$.

Problem 4. Consider the following description of the longitudinal field components of TM modes of the rectangular conducting waveguide (cross section dimensions $a \times b$):

$$\begin{cases} \mathbf{B}_z &= 0 \\ \mathbf{E}_z &= E_0 \sin(m\pi y/a)\sin(n\pi x/b)\hat{\mathbf{z}}. \end{cases}$$

(a) What are the restrictions on mode index (m, n)? Are these different from those of the TE modes?

(b) For $f = 40.1\,\text{GHz}$ light, how many (TE + TM) modes can propagate in total down a $3\,\text{cm} \times 1\,\text{cm}$ pipe? (you might wish to note the difference between f and ω). I suggest that you make a table of cutoff frequencies by m, n to assist in counting.

Solution 4.

(a) We've already shown that rectangular waveguides cannot support TEM modes, where $\mathbf{E}_z = \mathbf{B}_z = 0\hat{\mathbf{z}}$. Here, if either m or n are zero, we end up with a TEM mode, so the restriction is that neither m nor n may be zero. This is different from the TE restriction where one of m or n may be zero, but not both.

(b) Cutoff occurs when

$$f < \frac{c}{2}\sqrt{(m/a)^2 + (n/b)^2}.$$

Then for $f = 40.1\,\text{GHz}$, $a = 3\,\text{cm}$, and $b = 1\,\text{cm}$ we enumerate all (m, n) and check the expression above. I wrote a little python program to check these for me

```
import numpy as np
from scipy.constants import c

a = 0.03
b = 0.01
arr = np.ndarray(shape=(10, 10), dtype=int)
for m in range(10):
    for n in range(10):
        arr[m, n] = 40.1 * 10**9 < c * np.sqrt((m / a)**2 + (n / b)**2) / 2
print(arr)
```

and obtained output

$$\begin{pmatrix} 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix}$$

where the horizontal (column) index is n and the vertical (row) index is n. In the TE mode case, the entry where both m and n are zero isn't valid and so is discarded. In the TM case, all of the first row and column are discarded as neither m nor n may be zero. Since there are 23 total elements where $f < c\sqrt{(m/a)^2 + (n/b)^2}/2$, there are 22 allowed TE modes $(23 - 1)$ and 12 $(23 - 11)$ allowed TM modes, for a total of 34 modes which may propagate down the given waveguide.

Problem 5. Consider A hollow rectangular conducting waveguide of size $a \times b$ ($a > b$) which allows both TM and TE modes, with the propagation constant being given by:

$$k_{mn} = \sqrt{\left(\frac{\omega}{c}\right)^2 - \pi^2\left(\frac{m^2}{a^2} + \frac{n^2}{b^2}\right)}$$

This dispersion relation is sketched in the figure, below, for three modes; two of which have been labeled TE_{10} and TM_{11}.

(a) Use the figure to estimate [to a simple fraction (i.e. 3/2, 4/1, 6/5…)] the lowest possible aspect ratio of the waveguide (i.e. find a/b where a and b are integers below 10)

(b) Label the mode currently designated as "??" on the plot. If this mode is degenerate (either as TE and TM or in mn), make this clear.

(c) Estimate the penetration depth of $\omega = 0.5\,\mathrm{GHz}$ light down this hollow channel.

(d) At $\omega = 2.1\,\mathrm{GHz}$, which mode propagates with a larger phase velocity, TE_{10} or TM_{11}?

(e) $\omega = 2.1\,\mathrm{GHz}$, which mode propagates with a larger group velocity, TE_{10} or TM_{11}?

Solution 5.

(a) The figure indicates that $k_{10}(\omega \approx 1\,\mathrm{GHz}) = 0$ and $k_{11}(\omega \approx 2\,\mathrm{GHz}) = 0$. This is two equations in two unknowns,

$$0 = \sqrt{\left(\frac{10^9}{c}\right)^2 - \pi^2\left(\frac{1}{a^2}\right)}$$

and

$$0 = \sqrt{\left(\frac{2\cdot 10^9}{c}\right)^2 - \pi^2\left(\frac{1}{a^2} + \frac{1}{b^2}\right)}. \tag{1}$$

Squaring both of these and setting them equal yields

$$\frac{c\pi}{\sqrt{3}\cdot 10^9} = b \approx 0.54$$

and plugging that into **??**,

$$0 = \sqrt{\left(\frac{2\cdot 10^9}{c}\right)^2 - \pi^2\left(\frac{1}{a^2} + \left(\frac{10^9}{c\pi}\right)^2\right)}$$

which tells us that $a \approx 0.94$. The aspect ratio is then approximately 0.9/0.5 WHICH IS WRONG :(

(b) Label the mode currently designated as "??" on the plot. If this mode is degenerate (either as TE and TM or in mn), make this clear.

(c) Estimate the penetration depth of $\omega = 0.5\,\mathrm{GHz}$ light down this hollow channel.

(d) At $\omega = 2.1\,\mathrm{GHz}$, which mode propagates with a larger phase velocity, TE_{10} or TM_{11}?

(e) $\omega = 2.1\,\mathrm{GHz}$, which mode propagates with a larger group velocity, TE_{10} or TM_{11}?